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SumoRobot

by

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Mechatronic Project 478

Final Report

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Verklaring

Ek, die ondergetekende, verklaar hiermee dat die werk in hierdie verslag vervat, my eie oorspronklike werk is.

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EXECUTIVE SUMMARY

Title of Project
Design and Development of an Autonomous Mini Sumo Robot.
Objectives
Design and construct an autonomous Mini Sumo robot capable of detecting, pushing and removing an opponent from the sumo arena while remaining within the competition boundaries and complying with all competition constraints.
What is current practice and what are its limitations?
Current Mini Sumo robots primarily rely on simple sensor-based control systems for line detection and opponent detection. Their performance is often limited by restricted adaptability to different opponents, inefficient pushing strategies and the strict size, weight and power limitations imposed by competition rules.
What is new in this project?
This project investigates improved control strategies, sensor integration and mechanical design to enhance the robot's ability to detect opponents, remain within the arena and generate sufficient traction and pushing force during competition.
If the project is successful, how will it make a difference?
The completed robot will demonstrate a balanced integration of sensing, control and mechanical design for competitive Mini Sumo robotics. The resulting platform, control strategies and design methodology can serve as a reference for future robotics competitions and educational robotics projects. Several subsystems, including the control software and sensor integration techniques, can also be adapted for use in other mobile robotic applications.
What are the risks to the project being a success? Why is it expected to be successful?
Potential risks include unreliable sensor measurements, insufficient wheel traction or pushing force, and software or control issues affecting the robot's behaviour. These risks are mitigated through iterative testing, careful component selection, simulation, prototype evaluation and systematic software development throughout the project.
What contributions have/will other students make?
None. The project is conducted independently and does not directly build upon previous university robotics projects.
Which aspects of the project will carry on after completion and why?
The robot platform and control software can be further developed to improve autonomous strategies, sensing capabilities and mechanical performance for future robotics competitions or research projects.
What arrangements have been/will be made to expedite continuation?
Comprehensive documentation of the design process, hardware selection, software implementation, testing procedures and experimental results will be produced to assist future students in extending or improving the project.

Abstract

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Nomenklatuur

Konstantes

g =	$9.81\,\mathrm{m/s^2}$	

Veranderlikes

$Re_{\,\mathrm{D}}$	Reynolds getal t.o.v. deursnit	[ ]
x	Koordinaat	[m]
$\ddot{x}$	Versnelling	[$\mathrm{m/s^2}$]
θ	Rotasiehoek	[rad]
τ	Moment	[N·m]

Vektore en tensore

$\overrightarrow{\boldsymbol{v}}$	Fisiese vektor, sien vergelyking ...

Subscripts

a	Adiabaties
a	Koordinaat

Chapter 1

Introduction

*aiming for 2–3 pages

1.1 Background

The design of a mini sumobot is mainly limited by the competition guidelines and rules. This includes the physical dimensions of the robot, the characteristics and layout of the competition arena and the operational requirements and limits that must be satisfied during a match. The following section provides an overview of the regulations and rules according to Robogames. The Mini sumobot class is restricted to a maximum footprint of 100 mm x 100 mm with no height limit (RoboGames, 2026). The weight is limited to 500 g as well and the robot must be fully autonomous, with no remote control or human intervention allowed during a match.

The Robogames mini sumo competition arena, also known as the dohyo, is a circular platform with a diameter of 77 cm and a height of 2.5 cm. The surface is made of a smooth, black melamine coated wood and the boundary is marked by a white border with a width of 2.5 cm as seen in Figure 1.1.1. Inside this arena, two centerlines (also called shikiri-lines) are drawn to indicate the starting positions of the robots. The robots can be placed in any position behind the shikri-line (RoboGames, 2026).

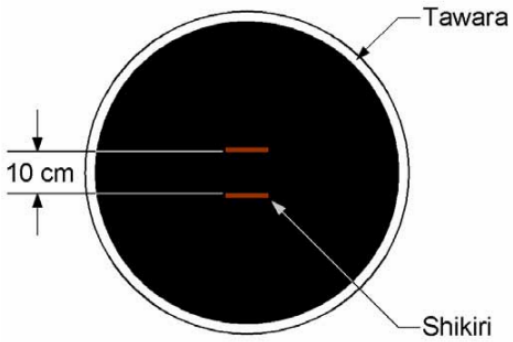


Figure 1.1.1: Standard mini sumo competition arena (Parallax Inc., 2005).

The main objective of these robots is to push the opponent out of the dohyo while avoiding being pushed out themselves. A 'Yuko' is scored when a robot is pushed out of the dohyo and the first robot to score two 'Yuko' wins the match. If both robots are pushed out of the dohyo simultaneously, the match is declared a draw and must be restarted (RoboGames, 2026). It should be noted that the use of any jammers or devices that could cause harm to the opponent is strictly prohibited. Using such devices could result in disqualification from the competition.

1.2 Objectives

The objective of this project is to design and develop a compact autonomous Mini Sumo robot that complies with the Robogames Mini Sumo competition regulations. The robot must be capable of detecting opponents while remaining within the arena boundaries through the implementation of a sensing and feedback system. A suitable control strategy will be developed to enable the robot to autonomously locate, engage and push its opponent out of the arena while avoiding the boundary line. Finally, the completed system will be validated through experimental testing and participation in the Mini Sumo knock-out competition to demonstrate its functionality and overall performance.

1.3 Motivation

Autonomous robotics combines theoretical knowledge, design and testing in a loop of trial, error and improvement. Such a seemingly simple project requires knowledge of a variety of engineering principles. It integrates mechanical design, the

design of electronic systems and the implementation of control and strategy algorithms.

All the above makes the Sumobot project particularly valuable for developing practical engineering and problem-solving skills. The design of a Mini Sumo robot requires careful consideration of traction, stability, sensing capabilities and control strategies to detect an opponent and successfully shove it out of the arena while remaining within the ring boundaries.

This project makes use of knowledge gained in the Computer systems, Electronics, Control systems, E-design and Machine Design modules. By integrating these areas, the project provides an opportunity to demonstrate the practical application of engineering principles in the design and development of a autonomous robotic system.

Chapter 2

Literature Review

*aiming for 8–10 pages

2.1 Autonomous Sumo Robots

I want to add in some form of background just for the flow to make more sense, but most of them fall under background/ the control strategy sections. Would briefly discussing the history of sumo robots be relevant? Most of what I could find just explained non-autonomous sumobots.

2.2 Detection and Sensing

The first step after starting the robot is to locate the opponent. Various sensor types can be used to detect the opponent, including infrared, ultrasonic, time-of-flight and vision-based sensors. The choice of sensor depends on factors such as detection range, response time, size, weight and power consumption.

Ultrasonic sensors measure distance by emitting a pulse of sound and measuring the time it takes for the echo to return. They are effective for detecting objects at medium range but may struggle with small or angled surfaces (Santos, 2013).

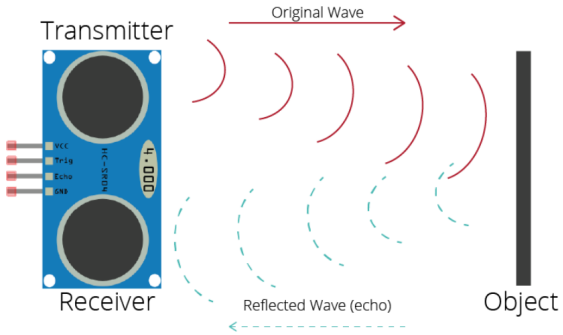


Figure 2.2.1: HC-SR04 ultrasonic sensor (Santos, 2013).

With the uncertainty of the opponent's design and possible need for accurately detecting angled surfaces, time-of-flight sensors may be a better option and are frequently used by competitive sumo robot teams. Time of flight sensors operate by emitting a light pulse and measuring the time it takes for the reflection to return. They are known for their fast response time and accuracy under different light conditions (ToF Sensors, 2026).

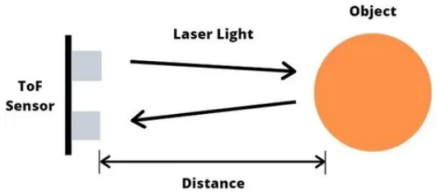


Figure 2.2.2: Time-of-Flight sensor (ToF Sensors, 2026).

IR sensors are used for opponent detection, but they can't measure distance as accurately as the other options. They work on the basis of detecting that there is something within its sensing range, but it cannot determine how far away the object is (Sisko, 2017).

Table 2.2.1: Comparison of opponent detection technologies

Parameter	**Ultrasonic HC-SR04**	**IR TCRT5000**	**ToF VL53L0X**
Maximum range	4 m	2 m	2 m
Minimum range	20 mm	0.2 mm	30 mm
Field of view	15°	24–35°	25°
Surface reflectivity sensitivity	Low	High	Moderate
Ambient light sensitivity	Low	High	Low
Typical measurement time	10 μs	10–15 μs	20–33 ms
Sensor interference	Moderate	Moderate	Low

2.3 Locomotion

2.3.1 Differential Drive Kinematics

Differential drive robots have two actuated wheels that operate on the same axis. These robots can be summarized by a state description consisting of both orientation and position (California Institute of Technology, 2018). This state can be modeled within the x-y plane as shown in Figure 2.3.1.

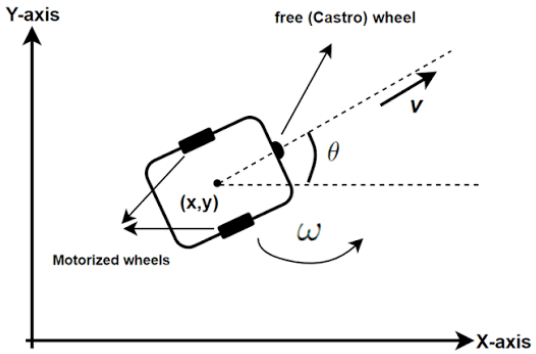


Figure 2.3.1: Differential drive robot kinematics (California Institute of Technology, 2018).

This figure represents a differential drive robot commonly used in 'Duckie Town' which includes a free castor wheel. However, these equations still apply to a differential drive robot without a castor wheel.

Assume the action vector of the robot is represented by $u = [u_r, u_l]$, where u_l and

u_r represent the left and right wheel velocities respectively. When both wheels rotate at the same angular velocity ($u_l = u_r > 0$), the differential-drive robot moves in a straight line along its current heading. The resulting linear velocity is directly proportional to the wheel radius, r. When the wheels rotate at equal speeds in opposite directions ($u_l = -u_r \neq 0$), the robot undergoes rotational motion about the midpoint of the axis of the wheels, producing no translational movement. From there on, the following equations can be used to determine the robot's position and orientation over time:

$$\dot{x} = \frac{r}{2}(u_r + u_l)cos(\theta) \tag{2.3.1}$$

$$\dot{y} = \frac{r}{2}(u_r + u_l)sin(\theta) \tag{2.3.2}$$

$$\dot{\theta} = \frac{r}{L}(u_r - u_l) \tag{2.3.3}$$

Where (x, y) is the position of the robot in the x-y plane, θ is the orientation of the robot, and L is the distance between the two wheels (Ras, 2025).

2.3.2 Wheel Material

2.3.2.1 Polyurethane

Polyurethane (PU) is commonly used for wheels because of its high abrasion and tear resistance. Commercially available polyurethane elastomers range from roughly 25–90 Shore A with densities of 1.24–1.27 g/cm^3 (Zenith Rubber Europe, 2021). Higher Shore hardness reduces deformation while increasing resistance to wear. This makes polyurethane suitable for applications requiring durable and high-load wheels.

2.3.2.2 Silicone Rubber

Silicone rubber is a softer elastomer that is also available for various Shore A hardness values. Silicone products are commonly available from 25–80 Shore A (Revchem Composites, 2022). Its lower hardness allows for greater conformity to the contact surface. This increases the contact area but also increase rolling deformation. While silicone rubber breaks down less easily than polyurethane, it is much more expensive. Additionally, it does not stick to surfaces during the moulding process and does not require a release agent.

2.3.2.3 Styrene-Ethylene-Butylene-Styrene

Styrene-Ethylene-Butylene-Styrene (SEBS) is a thermoplastic elastomer commonly used for wheels because it combines rubber-like flexibility with good abrasion and tear resistance. SEBS is available in a wide range of hardnesses, from approximately 10–80 Shore A, with densities ranging from roughly 0.87–0.93 g/cm^3 for unfilled grades (Mexpolimeros, 2025).

2.4 Control Strategies

2.4.1 Sumo Robot State Machine

The control strategy of sumorobots can ultimately be reduced to a simple state machine with four primary operating states:

- Start: waiting for the competition start signal and allowing the required delay period of 5 seconds (RoboGames, 2026).
- Search: rotating or moving to locate the opponent.
- Attack: driving towards the opponent while maintaining forward force.
- Back-off: reversing or turning when the dohyo boundary is detected.

2.4.2 Fuzzy Logic Control

Fuzzy logic control is a method that uses some degrees of freedom to determine the output rather than a binary true/false approach. It is particularly useful for systems with uncertainty or imprecision, such as those involving human-like decision-making or complex environmental interactions (Pantanetti *et al.*, 2025).

2.4.3 PID Control

Proportional-integral-derivative (PID) control is a feedback control method that adjusts a system's output based on the error between the desired and measured response. The three terms can be combined to form different controller types depending on the requirements of the system (MathWorks, 2026).

- **P:** Uses only the present error and is commonly used for simple systems where some steady-state error is acceptable.
- **PI:** Adds integral action to remove steady-state error and is commonly used for motor speed, temperature and flow control (MathWorks, 2026).

- **PD:** Adds derivative action to improve damping and reduce overshoot, and is commonly used for position and motion control where rapid response is required.
- **PID:** Combines all three terms to provide fast response while reducing steady-state error and overshoot. It is widely used in industrial and robotic control systems (MathWorks, 2026).

For a differential-drive sumo robot, these controllers can be applied to heading control using feedback from an IMU. The controller output can be used to adjust the relative speeds of the two motors, allowing the robot to maintain its desired heading.

2.5 Combat and Adaptive Strategies

-> flag servo motor strategy to confuse the opponent
-> oversaturation of sensors, ultrasonic and IR

2.6 Power Supply

The battery selection of an autonomous sumo robot affects its mass, available operating time and ability to supply the high currents required by the drive motors. Common battery technologies include lithium polymer (LiPo), lithium-ion (Li-ion) and alkaline batteries.

2.6.1 Lithium Polymer (LiPo) Batteries

LiPo batteries have a nominal cell voltage of 3.7 V and can provide high discharge currents due to their low internal resistance. Their specific energy is typically approximately 130–200 Wh/kg, while discharge ratings can exceed 20C for high-performance cells (Battery University, 2025). LiPo batteries are therefore commonly used in applications requiring a high power to battery mass ratio.

2.6.2 Lithium Ion (Li-ion) Batteries

Li-ion batteries have a nominal cell voltage of roughly 3.6–3.7 V and a specific energy of approximately 150–250 Wh/kg (Battery University, 2025). They generally provide higher energy density than LiPo cells but standard Li-ion cells have lower maximum discharge currents.

2.6.3 Alkaline Batteries

Alkaline batteries have a nominal voltage of 1.5 V per cell and a specific energy of approximately 80–110 Wh/kg (Battery University, 2025). Their internal resistance increases as the battery discharges, limiting their ability to provide sustained high currents. They can therefore be used for low-power electronics but are less suitable for applications requiring high instantaneous power from electric motors.

Table 2.6.1: Comparison of battery technologies

Parameter	**LiPo**	**Li-ion**	**Alkaline**
Nominal cell voltage	3.7 V	3.6–3.7 V	1.5 V
Specific energy	130–200 Wh/kg	150–250 Wh/kg	80–110 Wh/kg
Discharge capability	High	Moderate–High	Low
Internal resistance	Low	Low–Moderate	High
Rechargeable	Yes	Yes	No

2.7 Summary

Chapter 3

Specifications and Requirements

*aiming for 3–4 pages

3.1 Stakeholder Requirements

The stakeholder requirements for the autonomous sumorobot are listed in Table 3.1.1. The requirements are ranked in order of importance, with the most important requirement being ranked first. The primary stakeholders for this project include that staff and students who will be directly involved in the manufacturing, testing and the competition of the robot. The requirements are therefore based on the needs of these stakeholders, as well as the competition rules and regulations.

Table 3.1.1: Stakeholder requirements

SR ID	Description	Rank
SR1	The robot must comply with the Mini-Sumo competition rules.[1]	i
SR2	The robot must operate autonomously during the match.	ii
SR3	The robot must remain within the dohyo boundary.	iii
SR4	The robot must have enough traction and pushing force to compete against an opponent.	iv
SR5	The robot must withstand impacts or collisions and pushing forces without mechanical or electrical failure.	v
SR6	The robot must operate reliably throughout testing and competition without unexpected resets or system-related failures.	vi

Continued on next page

[1]This includes the applicable size, mass, starting delay, autonomous operation and competition requirements.

Table 3.1.1 continued from previous page

SR ID	Description	Rank
SR7	The robot must complete a full match using its contained power supply.	vii
SR8	The robot must be manufacturable using the available facilities, components and project budget.	viii

3.2 Functional Requirements

The functional requirements are derived from the stakeholder requirements and are listed in Table 3.2.1 below.

Table 3.2.1: Functional requirements for the autonomous sumorobot

FR ID	Description	Derived from
FR_01	The robot shall detect the opposing robot.	SR2
FR_02	The robot shall detect the edge of the dohyo.	SR3
FR_03	The robot shall move within the dohyo.	SR2
FR_04	The robot shall apply sufficient force to push the opposing robot out of the dohyo.	SR4
FR_05	The robot shall respond to the opposing robot in real time.	SR2, SR6
FR_06	The robot shall operate autonomously without human intervention after the start command.	SR1, SR2, SR3
FR_07	The robot shall have a maximum mass of 500 g and a maximum footprint of 100 mm x 100 mm.	SR1
FR_08	The robot shall be self-contained and shall not rely on external power, communication or physical support during a match.	SR1, SR7
FR_09	The robot shall begin operation only after the required competition start delay.	SR1
FR_10	The robot shall not damage the dohyo, the opposing robot or itself during normal operation.	SR1, SR5

Continued on next page

Table 3.2.1 continued from previous page

FR ID	Description	Derived from
FR_11	The robot shall be manufactured from materials and designed in such a way that it can withstand the forces and impacts experienced during a match.	SR5

3.3 Technical Performance Measures

The technical performance measures (TPMs) are derived from the functional requirements and are listed in Table 3.3.1. These performance measures will be used to evaluate and test the system.

Table 3.3.1: Technical performance measures for the autonomous sumorobot

TPM ID	Related FR	Description	Target	Range	Unit
TPM_01	FR_06	Start delay after power-on[1].	5	5–10	s
TPM_02	FR_01	Opponent detection time[8].	–	0–100	ms
TPM_03	FR_07	Robot footprint[2].	100×100	$\leq 100 \times 100$	mm
TPM_04	FR_07	Robot mass[2].	500	≤ 500	g
TPM_05	FR_07	Robot height[2].	40	< 100	mm
TPM_06	FR_02	Dohyo edge detection time[11].	20 ms	0–30	ms
TPM_07	FR_03	Maximum forward speed[9].	0.30	0.10–0.77	m/s
TPM_08	FR_04	Available wheel torque[3].	1.5	0.5–1.5	kgf·cm
TPM_09	FR_08	Nominal battery voltage[4].	12	7–14	V
TPM_10	FR_02	Edge detection distance from the boundary.	3	0–5	mm
TPM_11	FR_03	Turning radius.	0	0–50	mm

Continued on next page

Table 3.3.1 continued from previous page

TPM ID	Related FR	Description	Target	Range	Unit
TPM_12	FR_01	Opponent detection range[5].	770	0–770	mm
TPM_13	FR_01	Opponent detection field of view[5].	60	40–180	°
TPM_14	FR_03	Time to reach operating speed.	1	0.5–2	s
TPM_15	FR_04	Continuous pushing duration[7].	5	2–10	s
TPM_16	FR_08	Continuous operating time[6].	12	>3	min
TPM_19	FR_03	Steady-state speed error[10].	5	0–10	%
TPM_21	FR_03	Steady-state heading error[10].	0	0–10	°

[1] According to the Robogames Mini Sumo competition rules.
[2] Maximum robot dimensions and mass are specified by the Robogames Mini Sumo competition regulations.
[3] Wheel torque is specified per drive motor at the gearbox output shaft (see Appendix B).
[4] Values are based on the voltage requirements of most Sumobot peripherals. 7 V would be unideal, but manageable and 14 V would be manageable through step-down methods as well. Around 12 V would be ideal.
[5] The sensor must be able to detect the opponent within the specified arena (770 mm).
[6] Matches last for 3 minutes maximum according to the Robogames Mini Sumo competition rules. Ideally, the robot should be able to run for 3 consecutive matches without recharging or replacing the power source.
[7] Continuous pushing duration is based on the average time it takes to push an opponent out of the dohyo or to overcome wheel slip. The motor should not stall within this time.
[8] Estimate sensor processing time based on typical operation ratings for various sensors.
[9] Reaching this speed would allow the robot to cross roughly half of the dohyo in a second.
[10] These values are based on the expected performance of the robot during a match. Faster response is prioritized over perfect accuracy.
[11] At maximum speed of roughly 30 cm/s the robot will move 6 mm before detecting or reacting to the edge (20 ms x 30 cm/s = 6 mm).

Chapter 4

Concept Design

This section will discuss the concepts considered regarding the mechanical and electrical components of the autonomous sumorobot. Throughout this section a preliminary attack strategy was kept in mind to ensure that all decisions made would support that strategy instead of contradicting it. This preliminary attack strategy was to out-maneuver and flank the opponent rather than relying on pure pushing force.

4.1 Movement

4.1.1 Concept One: Four-Wheel Drive

The four-wheel drive concept uses four independently driven wheels with each wheel controlled by its own motor. Driving all four wheels increases traction, stability and pushing force while distributing the robot's weight more evenly across four contact points. The chassis should keep the motors aligned and position heavier components, such as the battery, low and near the centre to reduce the risk of tipping.

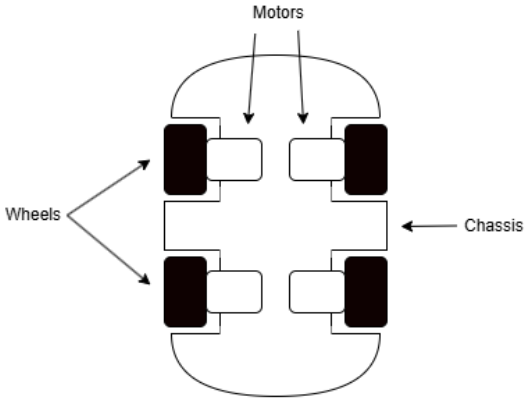


Figure 4.1.1: Four-wheel drive concept.

However, the additional motors increase mass, power consumption, wiring complexity and the space required within the 100 mm × 100 mm size limit. The four-wheel layout also experiences greater turning resistance due to wheel skidding. Overall, this concept prioritises traction, stability, and pushing performance (Arduino Forum, 2013).

4.1.2 Concept Two: Two-Wheel Drive

Differential drive uses two wheels, each driven by its own motor. This setup is commonly used to ease control of movement and turning as the robot can turn by varying the speed of each wheel. While this design won't reach the same level of traction and pushing force as a four-wheel drive, it is simpler to implement and is commonly used when the goal is to out-maneuver the opponent rather than out-torque them.

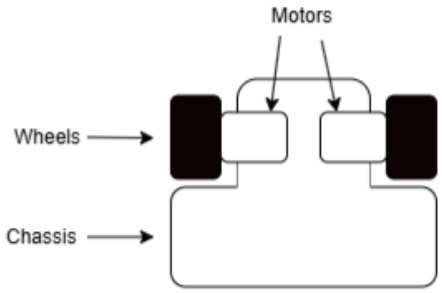


Figure 4.1.2: Two-wheel differential drive concept.

Differential drive is known for its zero turning radius (Sisko, 2017), which allows the robot to turn in place and quickly change direction. This is particularly useful in a sumo competition where the robot must react quickly to the opponent's movements and navigate within the limited space of the dohyo.

4.1.3 Concept Evaluation

The drive-train concepts were evaluated using criteria based on the functional requirements of the sumobot. The main criteria considered were manoeuvrability, traction, mechanical complexity and integration within the 100×100 mm size constraint. The decision matrix shown in Table 4.1.1 was used to compare these two concepts. The weights total up to 1 and the values are assigned on a scale of 0 to 2, where 0 is the worst and 2 is the best.

Table 4.1.1: Drive-train concept evaluation

Criterion	**Weight**	**Four-Wheel Drive**	**Two-Wheel Drive**
Manoeuvrability	0.35	1	2
Traction	0.35	2	1
Complexity	0.1	1	2
Integration	0.2	1	2
Weighted score		**1.35**	**1.65**

The differential-drive concept was selected with a score of 1.65 compared to the four-wheel drive's score of 1.35.

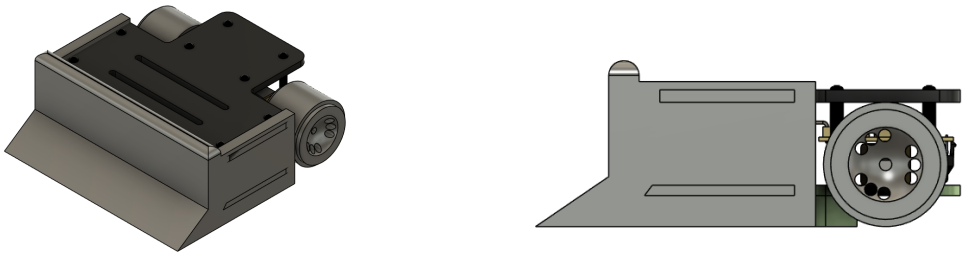
** Should I add a column that explains the reasoning or a column that links it to certain FRs? I originally removed it due to spacing/ the table spilling over the margins. I am planning to add a table like this to every section in this chapter. I added this as an example while I finish restructuring the flow of the other sections.

4.2 Engage opponent

In the mini-sumo class, most competitors use a wedge to engage the opponent. The wedge allows the robot to get underneath the opponent and lift it to minimize the opponent's traction.

4.2.1 Concept One: Flat Wedge

The first wedge concept considered was a flat wedge. This design is simple to manufacture and can be effective at lifting an opponent's wedge if the robot is able to get underneath it. Furthermore, this design allows for more space within the chassis for electrical components as well as room for sensor placement along the flat sides.

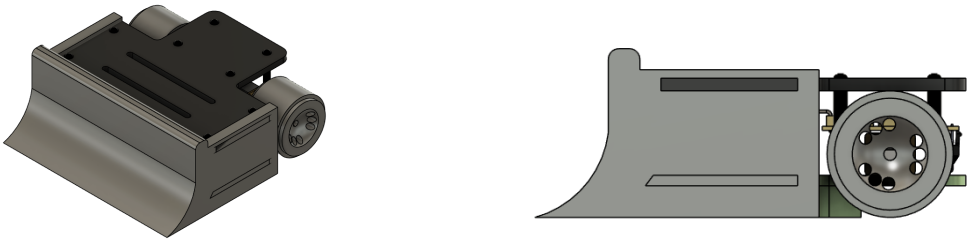


(a) Isometric view (b) Side view

Figure 4.2.1: Conceptual views of the flat-wedge design.

4.2.2 Concept Two: Curved Wedge

The curved wedge design is similar to the flat wedge, but with a curved front edge translating into a straight flat surface. This design is intended to help the robot redirect the opponent's wedge instead of lifting it and getting underneath the opponent.



(a) Isometric view (b) Side view

Figure 4.2.2: Conceptual views of the curved-wedge design.

4.2.3 Concept Three: Forked Wedge

The forked wedge design was considered to allow the robot to get underneath an opponent's wedge and lift it, while also providing a flat surface for sensor placement. The forked design allows the robot to straddle an opponent's wedge and lift it from both sides, potentially providing more stability during the lifting process.



(a) Isometric view (b) Front top-down view

Figure 4.2.3: Conceptual views of the forked-wedge design.

4.2.3.1 Concept Evaluation

4.3 Opponent and arena detection

For this section I am planning to redo all of the diagrams. They take up far too much space at the moment + I need to redo the text/ unit spacing. In my evalu-

ation table I will discuss which sensor is chosen for which purpose since the same sensor types were considered for arena and opponent detection.

4.3.0.1 Concept One: ToF Ultrasonic Sensors

The first concept considered was ultrasonic sensors - specifically the HC-SR04. These sensors are easy to implement both in hardware and software. Alongside that, they are relatively inexpensive compared to other sensors.

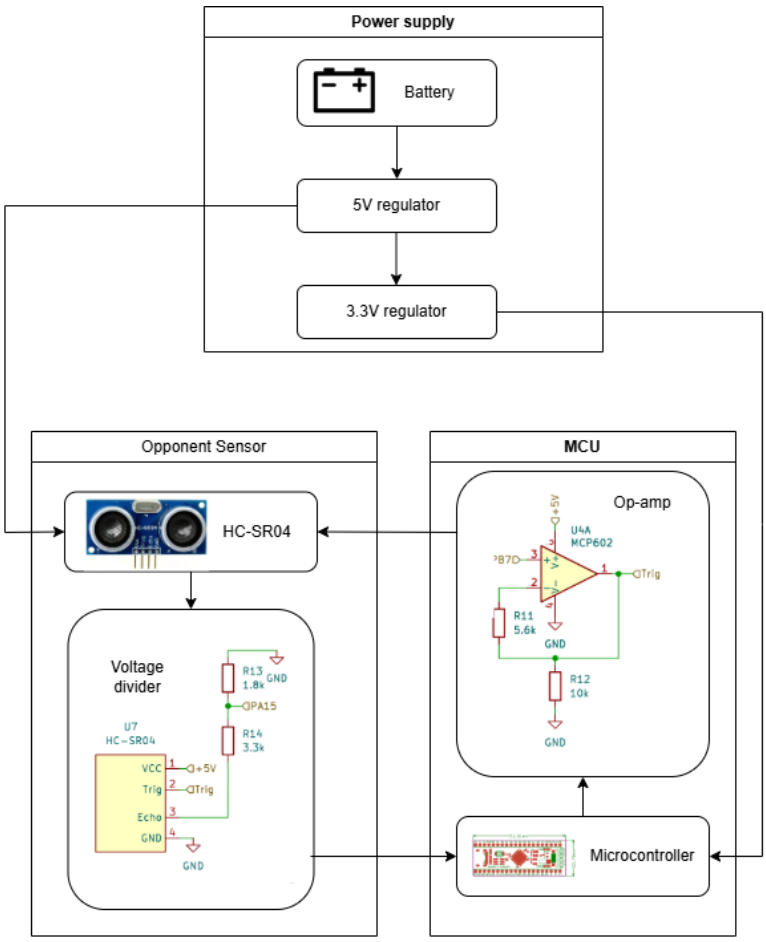


Figure 4.3.1: HC-SR04 ultrasonic sensor implementation.

These sensors work by sending out a pulse of ultrasonic sound and measuring the time it takes for the echo to return. The distance calculation can be implemented into the microcontroller's software through the following equation:

$$\text{Distance} = \frac{\text{Time} \times \text{Speed of Sound}}{2} \tag{4.3.1}$$

The ultrasonic sensor operates at a frequency of 40 kHz which is well below the required sensing latency of 100 ms (TPM_10) (ELECFreaks, n.d.). Alongside this, the sensor has a beam angle of 15–20 degrees and an operational range of 2 cm to 400 cm (ArcBotics, n.d.). Multiple sensors can be used to cover a wider area, allowing the robot to detect opponents from various angles. However, the sensor fails to detect objects that are too close, at an angle or that consists of a soft material. Furthermore, they operate at a voltage of 5 V, which is higher than the operating voltage of most micrcontrollers of interest. To accomodate this, voltage dividers and operational amplifiers would need to be implemented to ensure the sensor can be used properly. Overall, ultrasonic sensors are a good option for detecting opponents, but they have limitations that need to be considered when designing the robot's sensing system.

4.3.0.2 Concept Two: IR Sensors

Another sensor commonly used for object or obstacle detection is the infrared (IR) sensor. These sensors work by emitting an IR light and measuring the amount of light that is reflected back to the sensor. The distance to an object can be calculated based on the intensity of the reflected light.

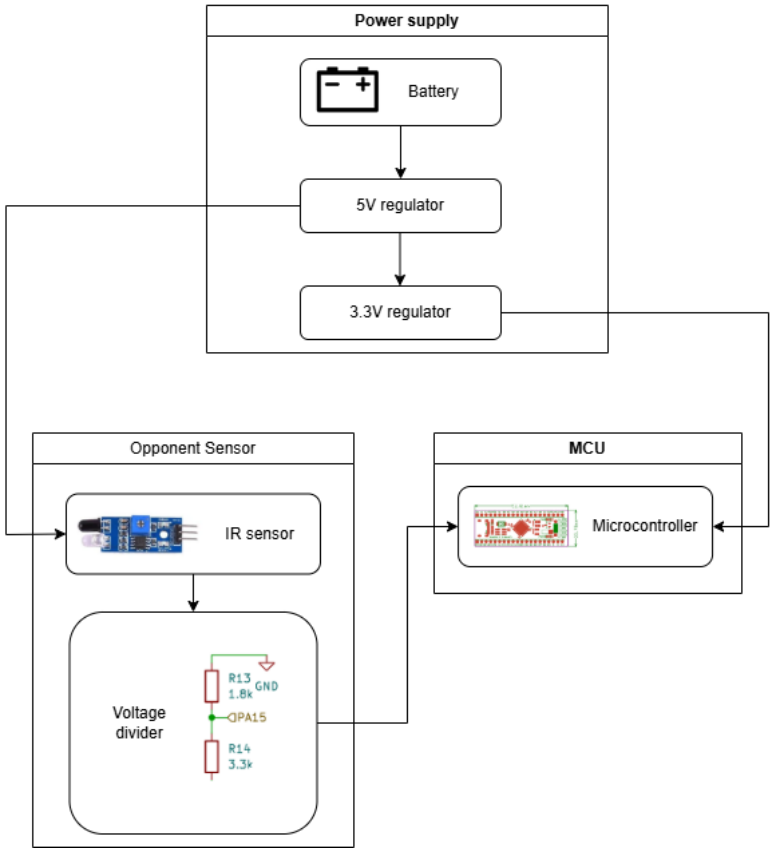


Figure 4.3.2: IR sensor implementation.

IR sensors operate between 3.6 V and 5 V (AM Electronics, n.d.). Similar to the HC-SR04, a voltage divider would need to be implemented to interface the sensor with a microcontroller operating at 3.3 V. The IR sensor has an effective range of 2 cm to 80 cm and has an on-board potentiometer to adjust the sensitivity of the sensor. The sensor has a beam angle of 35 degrees, which is slightly wider than the HC-SR04. However, the IR sensor is more susceptible to interference from ambient light and may not work well in bright environments.

4.3.0.3 Concept Three: ToF Laser Sensors

This sensor uses a laser to measure the distance to an object by calculating the time it takes for the laser light to reflect back to the sensor.

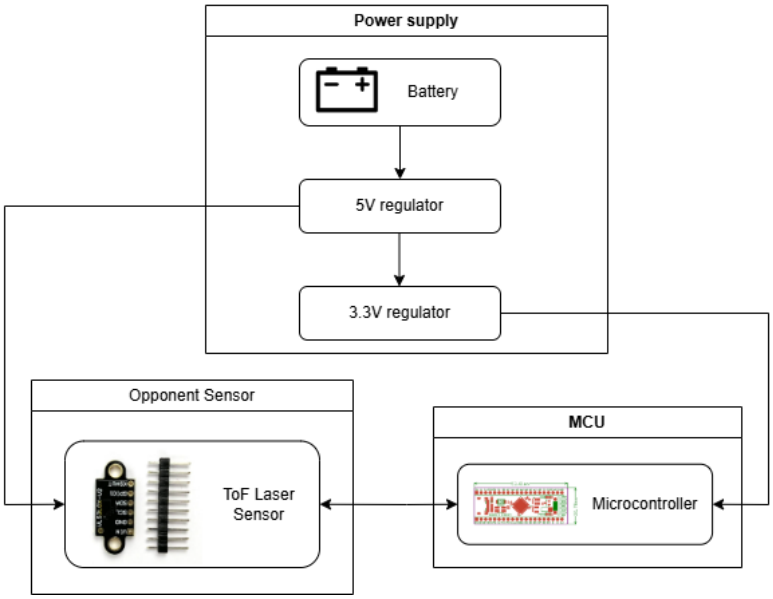


Figure 4.3.3: VL53L0X ToF laser sensor implementation.

These sensors can detect opponents at odd angles and are not as sensitive to ambient light as the IR sensors. The VL53L0X has a range of 30 mm to 2 m and an operating voltage of 2.6 V to 3.5 V (STMicroelectronics, 2024). The sensors operates at a frequency of 50 Hz, which is slower than the other sensors, but still well within the required sensing latency of 100 ms (TPM_10). Given the capability of operating at 3.3 V, the sensor can be directly interfaced with a microcontroller which would simplify the hardware design - especially if multiple sensors are involved. But the simpler hardware design comes at a cost - the VL53L0X is requires I2C communication to interface with the microcontroller. With multiple sensors involved, the registers of each sensor would need to be changed and the I2C bus would need to be managed properly to ensure that the sensors can be read correctly.

4.4 Control and measure motion

4.4.0.1 Concept One: Light-interrupter Wheel Encoders

Light-interrupter wheel encoders are a type of feedback system that can be used to measure the speed of the wheels. This is often useful for controlling the speed of the robot and ensuring that it is moving in a straight line. However, light-interrupter wheel encoders require a specific setup to work properly. In this case,

they would require a wheel with an extended shaft or a wheel radius large enough to house the encoder.

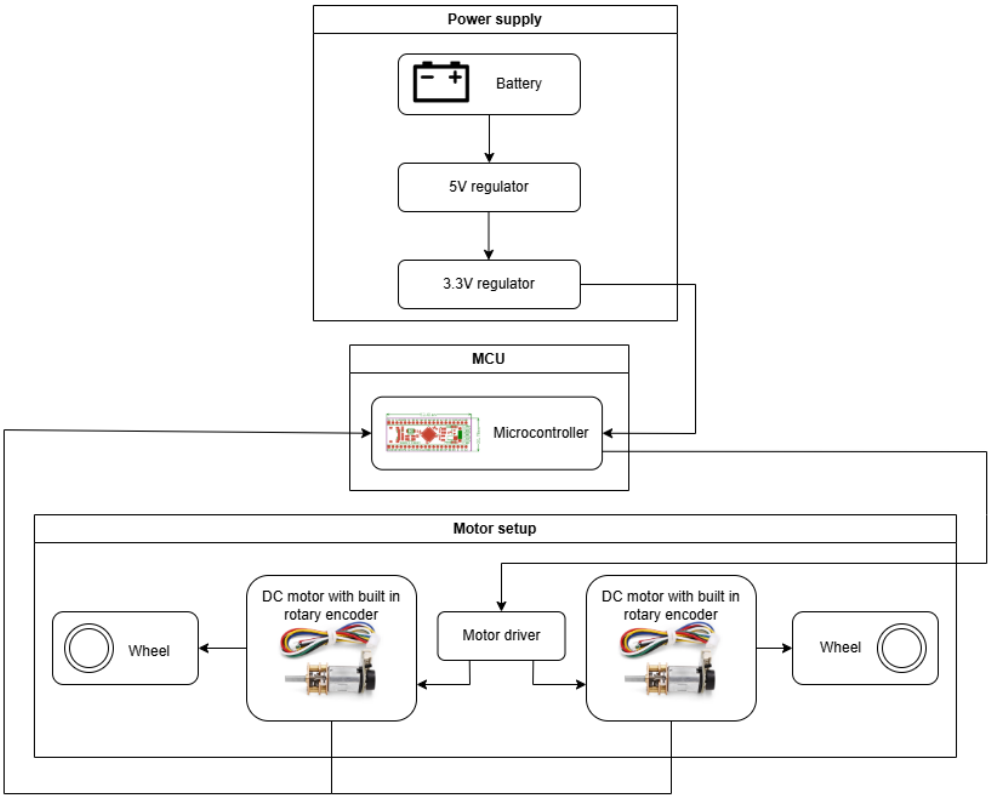


Figure 4.4.1: Light-interrupter wheel encoder implementation.

4.4.0.2 Concept Two: IR-sensor Wheel Encoders

Given the limitations regarding the availability of pre-mounted wheel encoders, the second concept considered was to use IR sensors to detect the rotation of the wheels. This would require a specific setup to work properly, but it would allow for more flexibility in the design of the robot. This concept would require a larger wheel radius to house the IR sensor - increasing speed and decreasing torque.

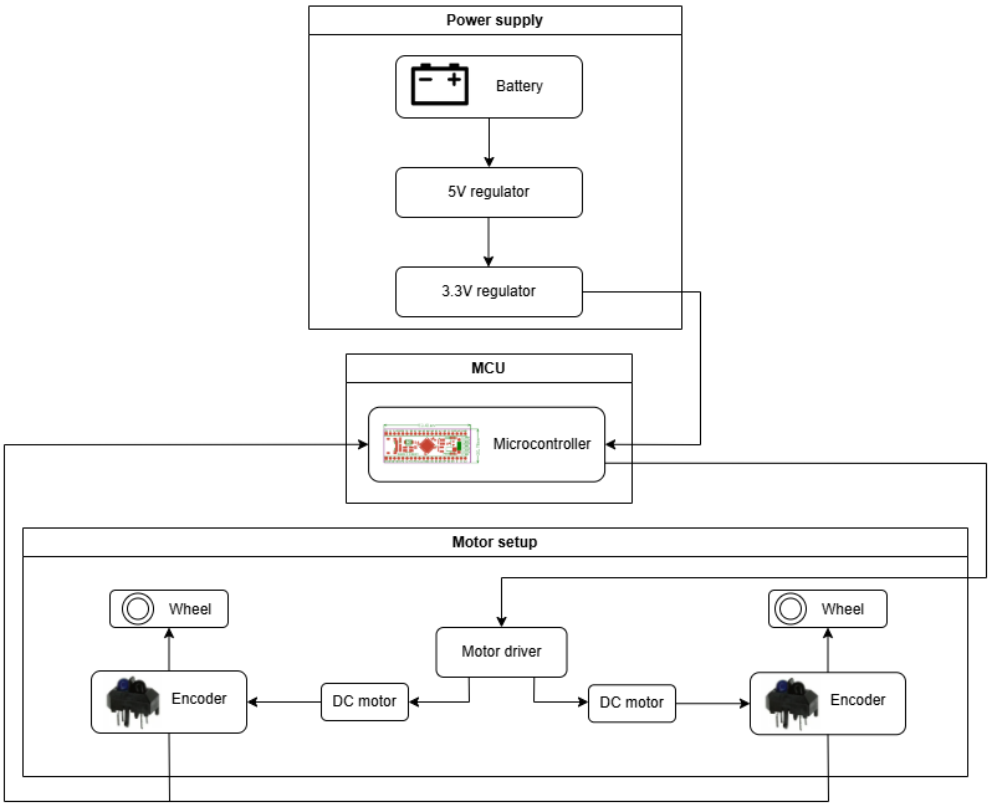


Figure 4.4.2: IR-sensor wheel encoder implementation.

4.4.1 Concept Three: MPU-6050 IMU with Potentiometers

In both of the previous concepts, only wheel speed could be measured. However, the final concept considered was to use an MPU-6050 IMU to measure the angular velocity of the robot as a whole. This would allow for more accurate control of the robot's movement and would not require any specific alongside the motors setup to work properly. However, the issue of varying wheel speeds would still need to be addressed. A common solution to this is to use potentiometers to manually tune and adjust the speed of each motor as needed.

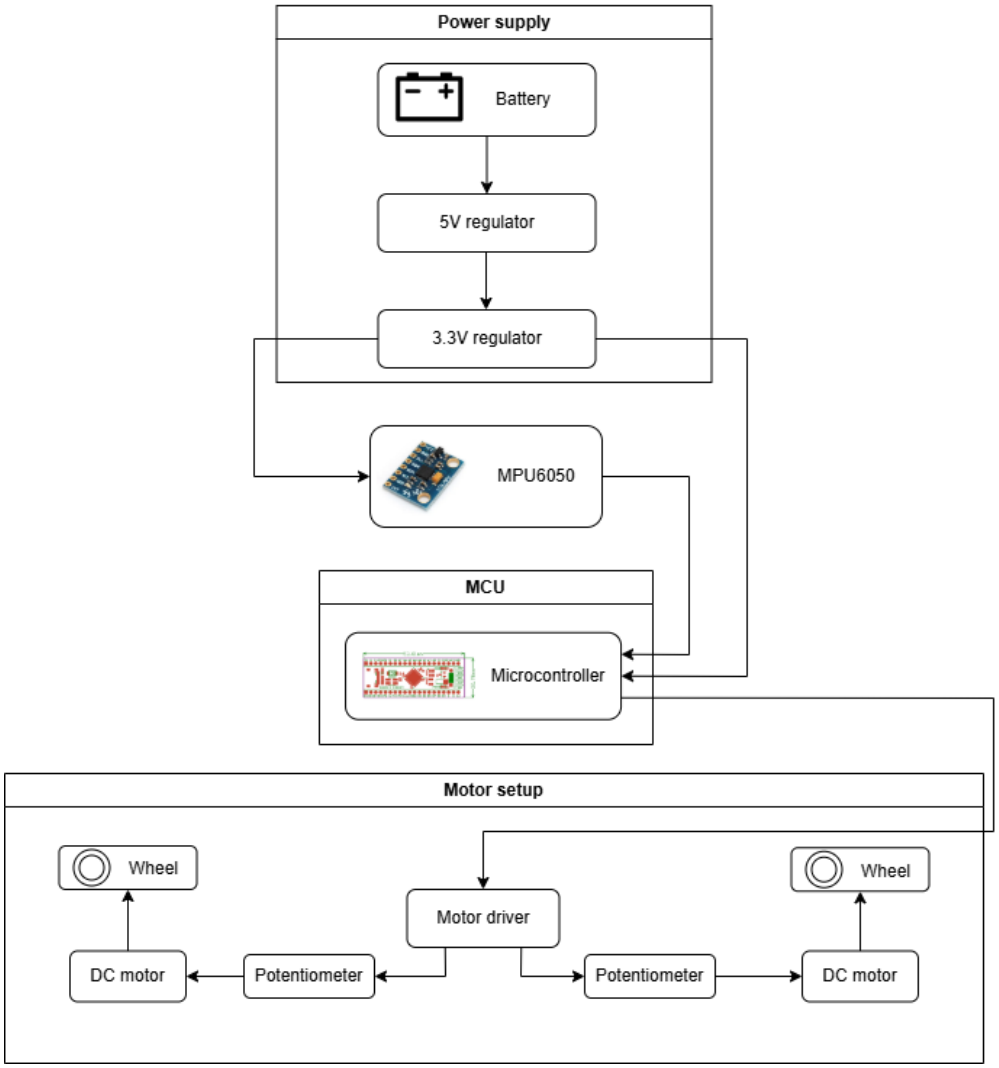


Figure 4.4.3: MPU-6050 IMU implementation.

This approach would then allow the system to meet FR_05 along with the sensors mentioned before.

4.4.1.1 Concept Evaluation

Chapter 5

Design Refinement

5.1 Electronic Components

5.1.1 Motor Selection

Might be a few mistakes here since I used 300 rpm instead of the original 500 rpm. I will update the calculations and go over the values again.
The motor selection process was by far one of the trickiest parts of the design process. The motors had to be small enough to fit within the 10 x 10 cm footprint alongside wheels, a battery and all other components. More importantly, the motors had to be locally available. Most small, high-torque motors are specifically designed for American or European Sumo-bot competitions. Local alternatives are either too weak or far too big. All options available would make it difficult to find a balance between torque and speed. The most suitable solution was to favour speed over torque as the aforementioned strategy was to out-manoever the robot rather than to rely on pushing power.

A 12V 300rpm N20 metal gear motor was selected as the most suitable option for the robot. The rated operating speed of 280rpm (Robo Factory, n.d.) would allow the robot to reach a top speed of 0.4 m/s - 0.5 m/s when paired with an outer wheel diameter of 35 mm (see Appendix B). This is still fast enough to reach one side of the arena from behind the shikri line within a second (theoretically meeting TPM_07). It should not be neccessary for the motor to reach this speed during most of the match, as the robot will be moving at a slower speed to avoid overshooting the opponent or the edge of the arena. The trade-off to meet this speed is that the torque of the motor is only 0.8 kgfcm per motor. This would result in a rated pushing force of 5N from both motors. However, the stall torque of the motor is 4 kgfcm per motor (Robo Factory, n.d.). While designing around stall torque is not ideal, these motors do still offer more than enough headroom to push more momentarily.

5.1.2 Motor Driver Selection

A TB67H420FTG Dual motor driver carrier was selected to control the motors. This motor driver is capable of driving two DC motors with a supply voltage of 10V to 47V and is rated to supply 1.7 A continuously per motor. This makes it suitable for the 12 V N20 motors which are rated to draw 1.5 A at stall. The TB67H420FTG motor is an H-bridge driver IC that allows bidirectional control. Furthermore, it uses MOSFETS whereas older motordrivers utilize Bipolar Junction Transistors (BJTs). MOSFETS are more efficient than BJTs, which means that they can handle higher currents and dissipate less heat. The carrier has built-in protection against under-voltage, over-current, and over-temperature conditions (Micro Robotics, 2026*b*).

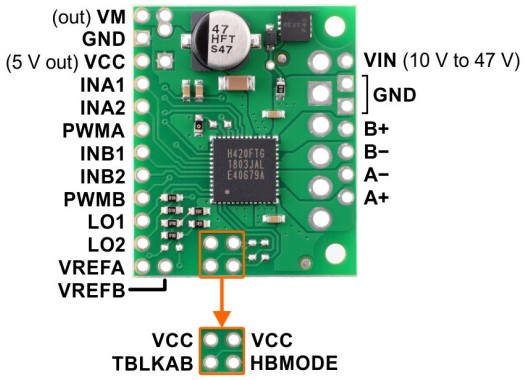


Figure 5.1.1: TB67H420FTG Dual Motor Driver Carrier (Micro Robotics, 2026*b*)

I'm not sure if it might be worth adding MOSFET vs BJT information in the literature review. I found it interesting for motor application, but it might not be too relevant to sumobots in general.

5.1.3 Microcontroller Selection

The microcontroller is the central processing unit of the autonomous sumorobot. It is responsible for executing the control algorithms, processing sensor data and controlling the motors.

Given its dependence on all other peripherals, the microcontroller could only be selected towards the end of the design process. During the selection process, there were several limiting factors. The microcontroller had to fit within the 10cm×10cm footprint of the robot while providing sufficient processing capability and enough

GPIO pins with the required functions to interface with all of the selected peripherals. The following microcontrollers were considered:

Table 5.1.1: Comparison of microcontrollers

Specification	**ESP32 DevKit**	**Arduino Nano**	**STM32F411 Blackpill**
Processor	240 MHz	16 MHz	100 MHz
Flash	4 MB	32 KB	512 KB
SRAM	520 KB	2 KB	128 KB
GPIO	25	22	30+
PWM	16 channels	6 channels	Multiple
I^2C	2	1	3
Logic	3.3 V	5 V	3.3 V
Board Size	51 × 28 mm	45 × 18 mm	52 × 25 mm

The ESP32 DevKit V1 was selected as the most suitable option for the robot. Its relatively small development-board footprint, large number of configurable GPIO pins and support for common communication protocols such as I^2C and SPI allow it to interface with all of the required peripherals. The ESP32 provides sufficient processing and peripheral capabilities for the autonomous sumorobot, with a dual-core 32-bit processor operating at up to 240 MHz, 4 MB of flash memory and hardware PWM for independent motor control. These capabilities allow the ESP32 to process sensor data and execute the required control algorithms in real time.

However, it should be noted that ESP32s have relatively noisy ADCs which can lead to inaccurate readings. Referring back to Section 4.3, it was ensured that all sensors either relied on I^2C or digital GPIOs to avoid the use of the ESP32's ADCs. Table 5.1.2 shows the pins needed for each peripheral which aids in the planning of the pin configuration for the microcontroller.

Table 5.1.2: Pin planning for microcontroller

Peripheral	Pin Type	Number of Pins
VL53L0X Time-of-Flight Sensors	I2C SDA	1
	I2C SCL	1
	GPIO	3–5
IR Edge Detection Sensors	GPIO	3
MPU-6050 IMU	I2C SDA	1
	I2C SCL	1
Motor Drivers	PWM	2
	GPIO	4
Push Button	GPIO	1
DIP Switch	GPIO	4

5.1.4 Power Supply

When choosing a power supply, it is important to consider the voltage and current requirements of the entire system. The power supply must be able to provide sufficient voltage and current to all components, including the microcontroller, sensors, motors and any other peripherals.

Lithium polymer (LiPo) batteries are a popular choice for portable electronic devices due to their high energy density and lightweight design. Alongside all of the upsides for using LiPo batteries, it should be noted that they require careful handling and charging to avoid safety hazards. LiPo batteries can be dangerous if punctured, overcharged, or exposed to high temperatures (ChinaHobbyLine, 2024).

With all this in mind, a 3S 11.1V LiPo battery with a capacity of 750mAh and a discharge rate of 45C was selected as the power supply for the system. A datasheet could not be found for this specific component - instead, the specifications provided by the supplier were used to determine the suitability of the battery for the system.

Given the 750mAh and 45C rating of the battery, the maximum current that can be drawn from the battery is 33.75A. Using this and using conservative estimates for all present electrical components, the total current draw of the system was calculated to be 4.4A. This offers a safety margin nearly 8, which is more than sufficient to ensure that the battery can provide enough current to all components

without being overdrawn.

5.1.5 Voltage Regulator and Circuit Protection

Voltage regulators are used to maintain a constant output voltage regardless of changes in input voltage or load conditions. They ensure that the microcontroller and other sensitive components receive a safe and stable voltage supply. The voltage supply was first stepped down to 5 V before being further stepped down to 3.3 V for the microcontroller and other components. Given the current draw of the system, a LM2596 DC-DC buck converter was selected as the voltage regulator for the system. The LM2596 is a popular choice for voltage regulation due to its high efficiency, low dropout voltage, and wide input voltage range. The LM2596 can handle a maximum input voltage of 18 V and can provide a regulated output voltage of 1-17 V with an adjustable potentiometer. The LM2596 can provide a maximum output current of 2 A (Micro Robotics, 2026*a*), which is sufficient for the current requirements of the sensors and other peripherals (see Appendix B).

While the voltage regulator is the biggest source of safety for the system regarding voltage fluctations, current protection is also important. A 200 - 300 mA fuse will be installed in series with the power supply to the microcontroller to protect all important components from overcurrent situations. Ideally, the microcontroller should not pull more than 10 - 20 mA while its GPIO pins are rated for 8 mA (STMicroelectronics, 2025).

5.2 Mechanical Components

5.2.1 Sensor Placement

** Should I rather move this section to the testing section and refer back to it here?
A preliminary test was performed regarding the sesnor placement and what their optimal placement or orientation might be. The goal was to detect as much of the arena while using as little sensors as possible. Figure 5.2.1 shows the results of the sensor readings and what the actual detection range of the VL53L0X sensors being used is. See Appendix X for the test setup and procedure. I need to update the Appendix hence the placeholder Appendix X.

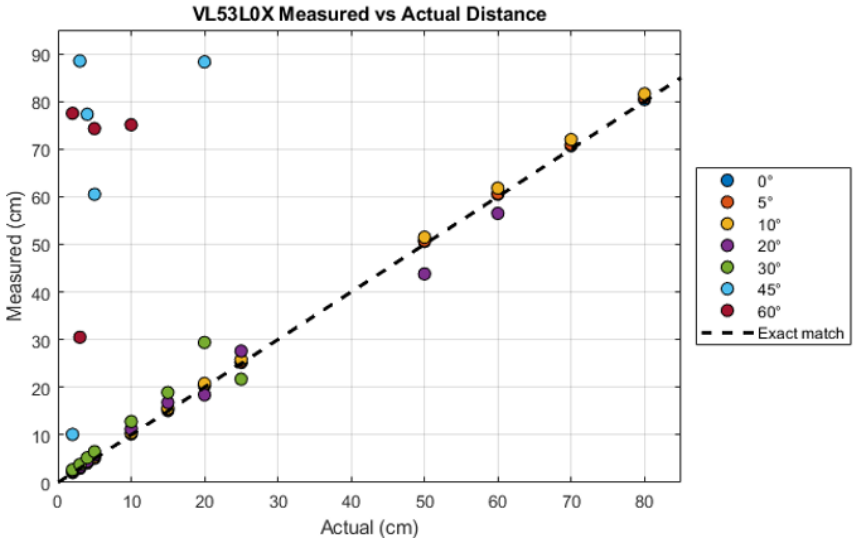


Figure 5.2.1: Sensor reading test

The sensors stopped detecting the test object at an angle of roughly 20–30°. After that, the sensors read the distance of a wall in the background - grouping the readings around 80 cm. This angle was used to plot the different sensor placements in MATLAB - simulating what placement would be best to detect the most of the arena.

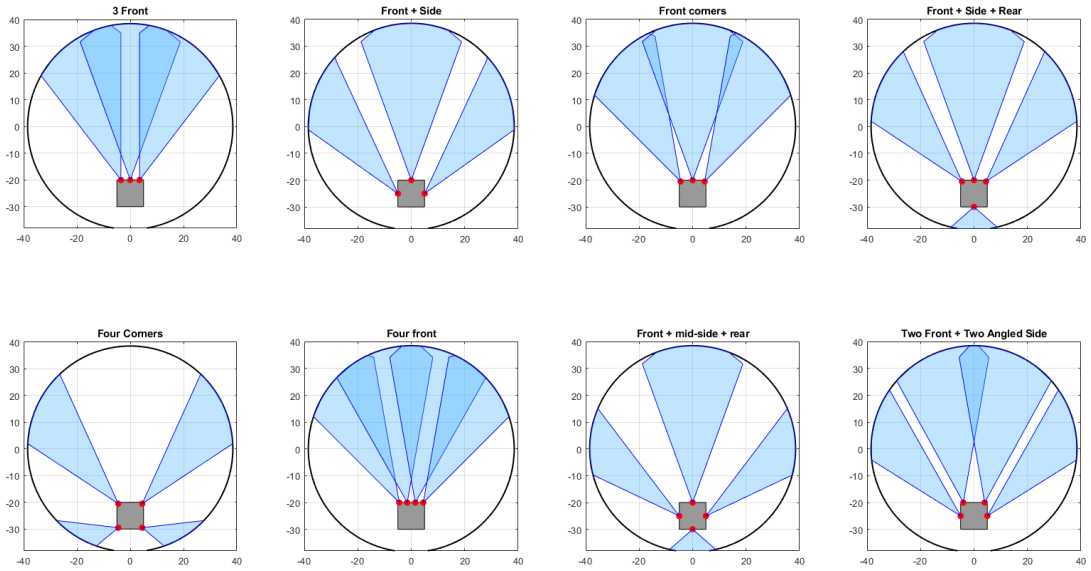


Figure 5.2.2: Sensor placement simulation

Two front facing sensors paired with two angled side-facing sensors were selected as the optimal placement for the robot.

5.2.2 Wedge

While a curved wedge was chosen during the concept development stage, it still needed to be refined to interface with all other components such as the chassis and sensor mounts. A key consideration throughout the entire mechanical design process was the screw placement. Bots of this size usually won't place much strain on the screws, but it is still good practice to keep in mind which forces would be acting on which surfaces. For example, the screws of the wedge were placed forward facing to ensure that the majority of the anticipated collission-related forces would be applied in-line with the screws - avoiding shearing forces.

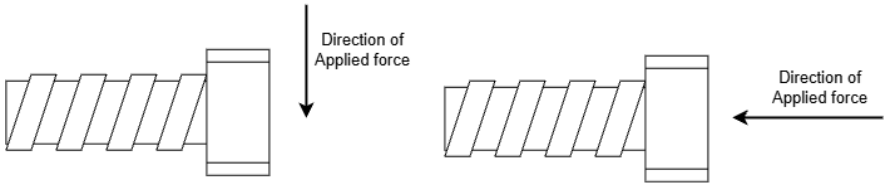
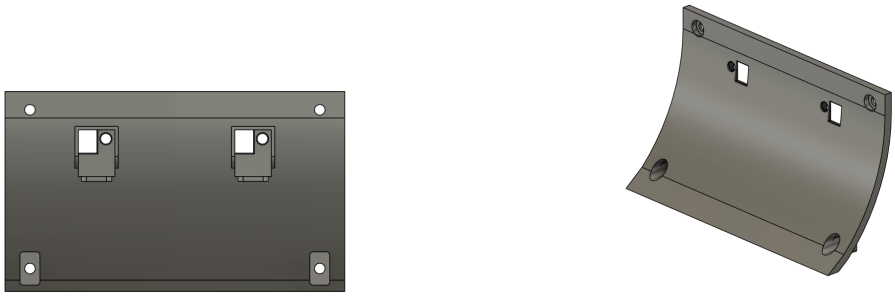


Figure 5.2.3: Anticipated forces on the screws.

Figure 5.2.4 shows the final wedge design with the screw placement and the cutouts for the sensors.



(a) Back of the final wedge.

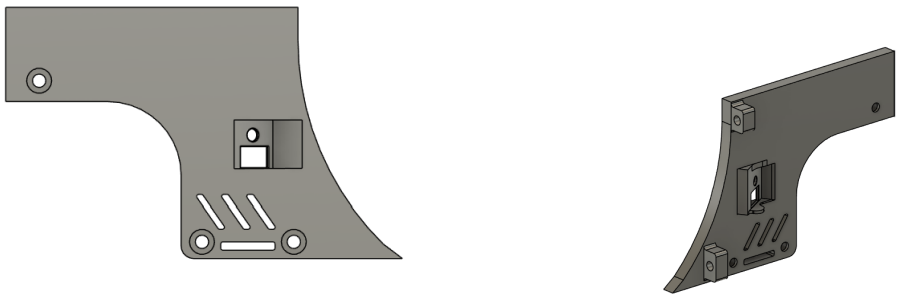
(b) Isometric view of the final wedge.

Figure 5.2.4: Final front wedge design.

*I still want to include a diagram showing the forces acting on the wedge

5.2.3 Side Frame

The sides are mainly there to provide a mounting surface for the sensors and to protect the electronics from collisions. The sides were designed to mount one VL53L0X sensor alongside a small bracket that mounts the IR edge detection sensor. Figure 5.2.4 shows the final wedge design with the screw placement and the cutouts for the sensors.



(a) Front view of the side. (b) Isometric view of the final side panels.

Figure 5.2.5: Final side panel design.

The gaps near the lower side allow for more natural light to reach the IR edge detection sensor.

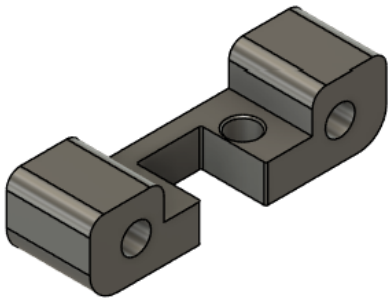
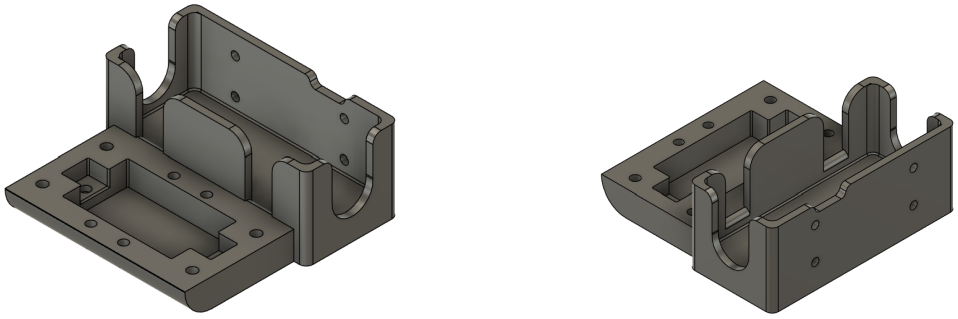


Figure 5.2.6: IR edge detection sensor mounting.

5.2.4 Chassis

The chassis consists of mountint space for the motors, a bracket to protect the battery and a slot to place extra weights withing. This slot sits directly under

the drive line - allowing the robot to have a lower center of gravity and more stability. Figure 5.2.4 shows the final wedge design with the screw placement and the cutouts for the sensors.



(a) Isometric front view of the final chassis. (b) Isometric back view of the chassis.

Figure 5.2.7: Final chassis design.

5.2.5 Electronic mounts

I made a few brackets to keep the electronics organized. It wasn't apart of the planning and I could only do it once my circuit was built. Is it worth discussing here?

5.2.6 Wheel Selection

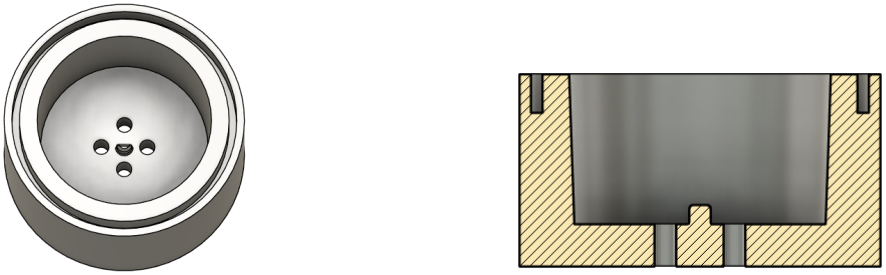
Three commonly used materials for bot-specific wheels were considered and discussed in Section 2.3.2.

Table 5.2.1: Comparison of wheel materials

Material	**Shore A**	**Density (g/cm^3)**	**Friction coefficient**
Polyurethane	25–90	1.24–1.27	0.5–1.2
Silicone rubber	25–80	1.21	0.4–0.8
SEBS	10–80	0.87–0.93	0.4–1.0

Moulding wheels from a high-friction material such as polyurethane is common practice amongst competitive robotics. Coupled with a larger width, this would provide the robot with a larger contact patch and far better traction than any off

the shelf wheels (SEBS). To achieve this, a custom wheel mould and hubs were designed.



(a) Wheel mould (b) Wheel mould draft angle

Figure 5.2.8: Conceptual views of the wheel mould designs.

Shore 40 A Vytaflex was selected specifically for its hardness rating. However, this material fuses easily with PETG. While it is ideal for the polyurethane to fuse to the wheel hub, a release agent paired with a draft angle of 2°was used to ensure that the polyurethane would not fuse to the mould cavity. Initially, there were some concerns about keeping the hub in place or preventing it from floating. The density of PETG is 1.25 g/cm^3 while the density of Vytaflex is 1.15 g/cm^3. Since the density of PLA is higher than that of Vytaflex, the hub would sink into the mold cavity and floating should theoretically not be an issue.

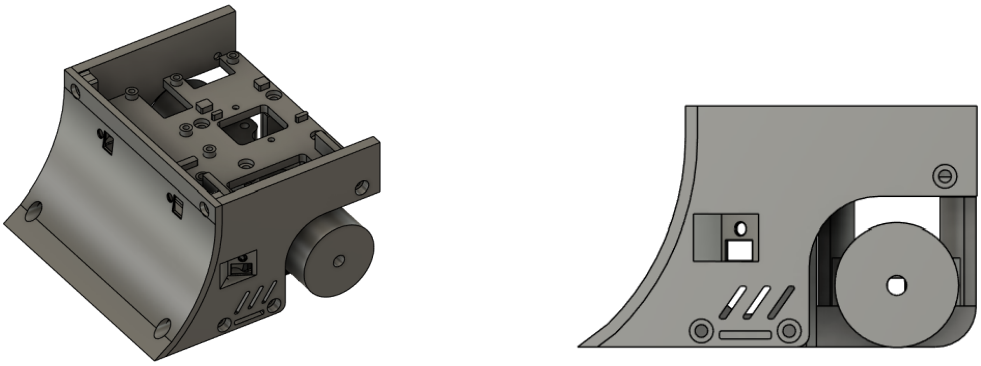


(a) Wheel hub (b) Wheel hub top view

Figure 5.2.9: Conceptual views of the wheel hub designs.

The wheel hub was designed in such a way that the polyurethane would slot into the six outer gaps - making it more difficult to release or shear when the robot is in motion.Furthermore, the center slot was designed with two extra gaps around the arched side of the D-slot to allow the N20 motor shaft to be inserted into the hub more easily. These gaps create a slight relief area in the material while the flat side remains in contact with the motor shaft.

5.3 Final Assembly



(a) Front view. (b) Side view.

Figure 5.3.1: Final CAD assembly.

***Should I include the full circuit schematic plan here as well or just leave it in the appendix? I thought it would fit nicely to conclude this chapter, but it is incredibly cluttered and hard to read on such a small scale.
-> Alternatively, flow diagram showing the morphological flow of the design + which TPMs are linked to which element.

Chapter 6

Software Design and implementation

* This section isn't finalized. I noted some details down that will be discussed (I'll fix the first person writing as well)

6.1 Startup Sequence

I implemented a hardware check. Since some components froze the system and I wanted a direct indication of what was going on, I used the onboard LED to indicate what failed.
-> ToF sensors, one pulse for first sensor left side, increasing to 4 pulses for the right side sensor. (before button press, on startup)
-> IMU, rapid blinking if not detected (after button has been pressed)
-> IMU int2 failure, three rapid flashes, stop, three rapid flashes, stop (after button press)
-> All good - LED off before button press, LED on after button press with 5s delay
-> IMU is reinitialized or configured as a 'new zero' whenever the robot is turned on. Wasn't working consistently before this.

6.2 Search Algorithm

-> spinning until detected. Sometimes it missed the object so if nothing is detected after two spins it now roams the arena until it detects an object.
-> in the case of sensor oversaturation or unreliable readings (comparing the two front sensors), the robot will roam as well until impact is detected via the IMU
-> IMU still needs to be tuned a bit, picked up false positives/ random impacts. Should just be a threshold change that will be discussed here.
-> Fuzzy logic states

6.3 Attack Algorithm

-> opponent detected
-> motors boosted full speed
-> flanking path follow + PID (ended up using PI for one set of wheels, but PD for a better set of wheels)
-> Once impact detected, PID is disabled and the robot will continue to push
-> If not movement is detected, but it is in this pushing state, the robot will back off and attempt the wood-pecker approach to avoid stalling the motors
-> Fuzzy logic states

6.4 Back-off Algorithm

-> edge detection, overrides attack and search

6.5 Feedback and Control Systems

-> will discuss the PID equation and its implementation here
-> error = yaw vs expected yaw
-> the actual PID values will be discussed in the testing section.

Chapter 7

Testing and Refinement

7.1 Power Supply

The original design included a DC-to-DC buck converter to step down the voltage and regulate 5 V and 3.3 V voltage lines. While these converters are efficient and produce less heat than linear regulators, the inexpensive options are prone to voltage shifts and spikes. Especially LM2596 which has an onboard potentiometer to adjust the output. Whenever the board was bumped or moved, the output voltage would shift. This would especially be an issue given the expected impacts the sumo robot would experience during a match.

These converters were ultimately replaced with linear regulators. A L78C5 and LD1117 were used to regulate the 5 V and 3.3 V lines respectively. The circuits were built using an 1N4007 diode to protect against reverse voltage and 100 nF as well as 10 µFcapacitors to reduce noise. The same power indicating LED was connected to the 5 V line to indicate when the board was powered. Figure 7.1.1 shows the final power supply schematic.

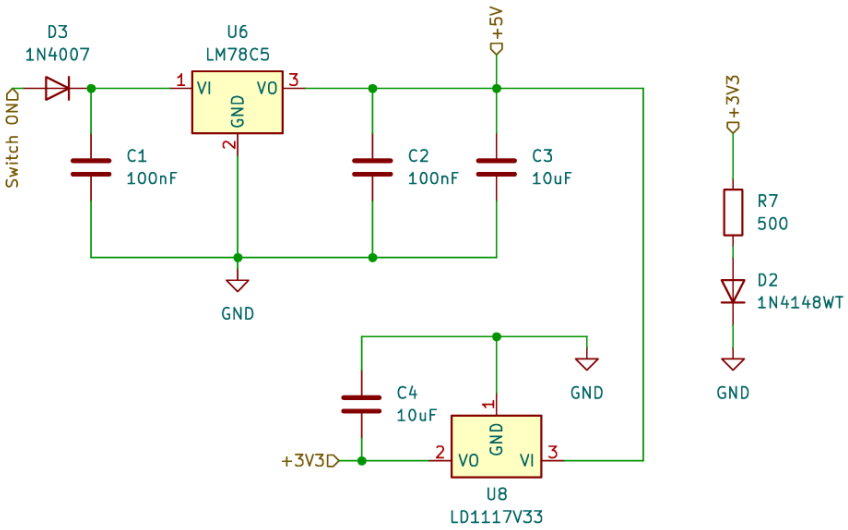


Figure 7.1.1: Power Supply Schematic

7.2 ToF Sensor

-> originally out by a constant of 2 cm, calibrated in code

7.3 IR Sensor

-> edge detection/ response testing discussion here

7.4 PID Tuning

-> ended up with Kp=18, Kd=0.5 for faster response

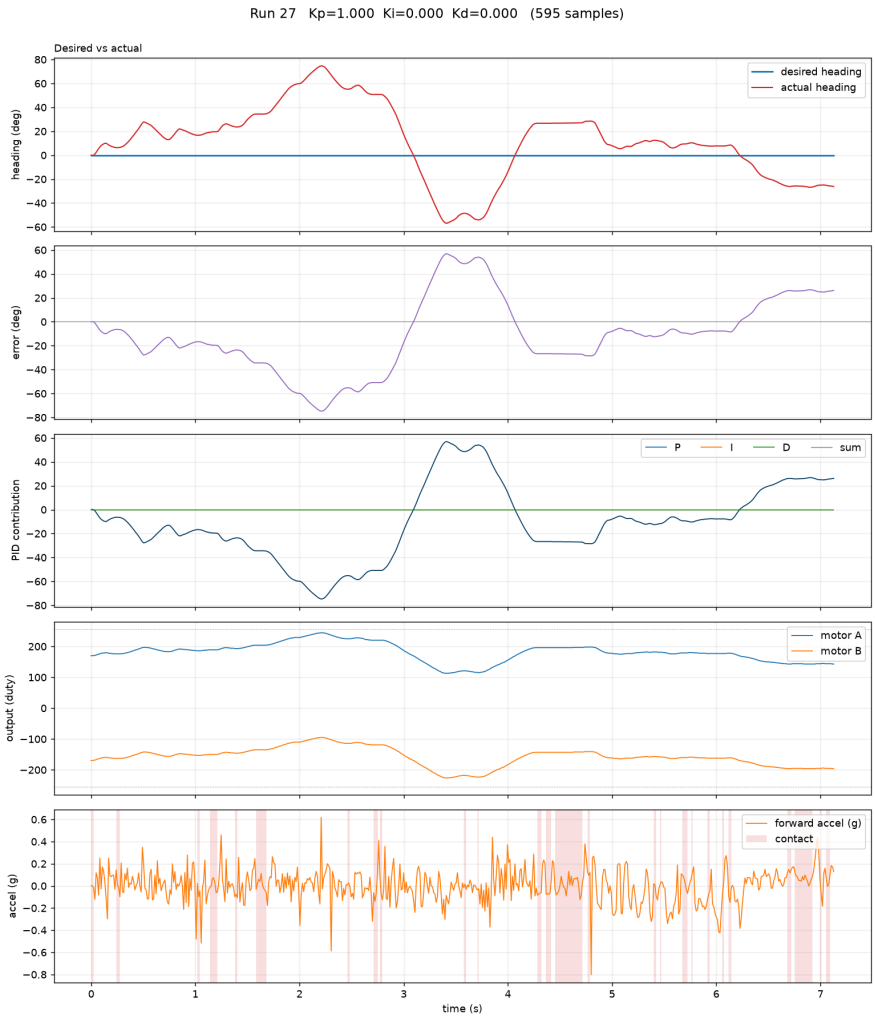


Figure 7.4.1: One of the first runs

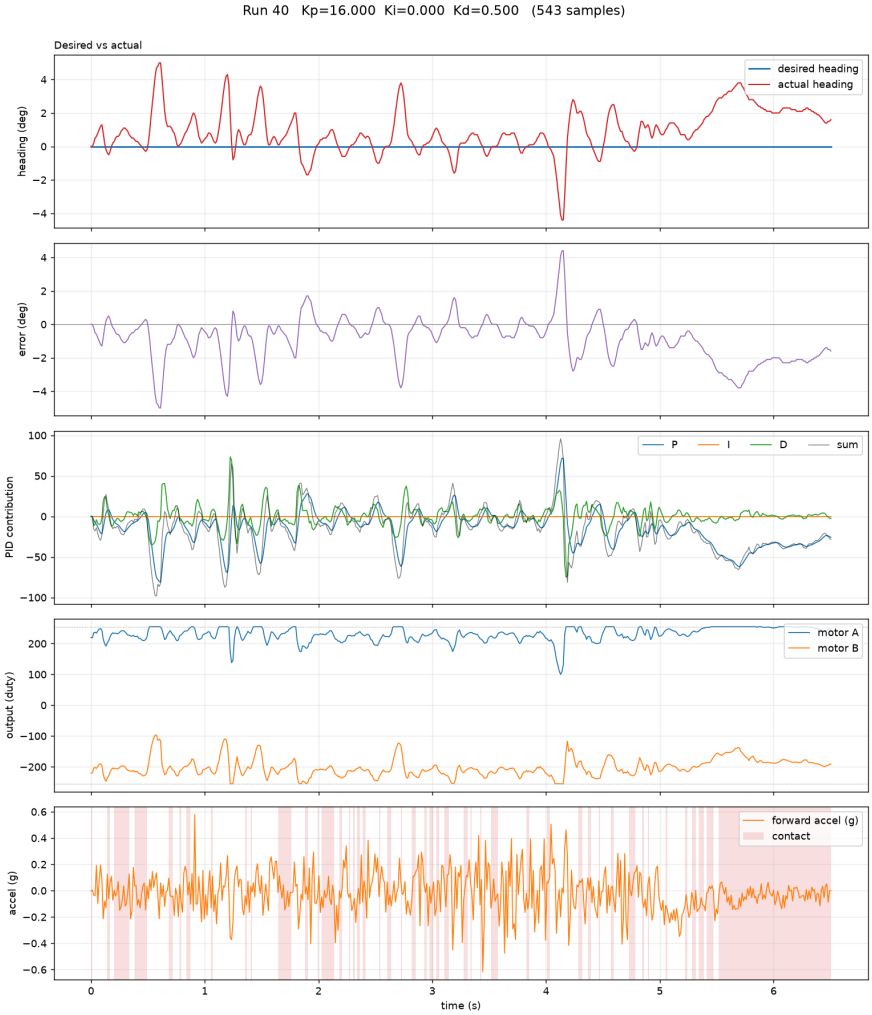


Figure 7.4.2: In between

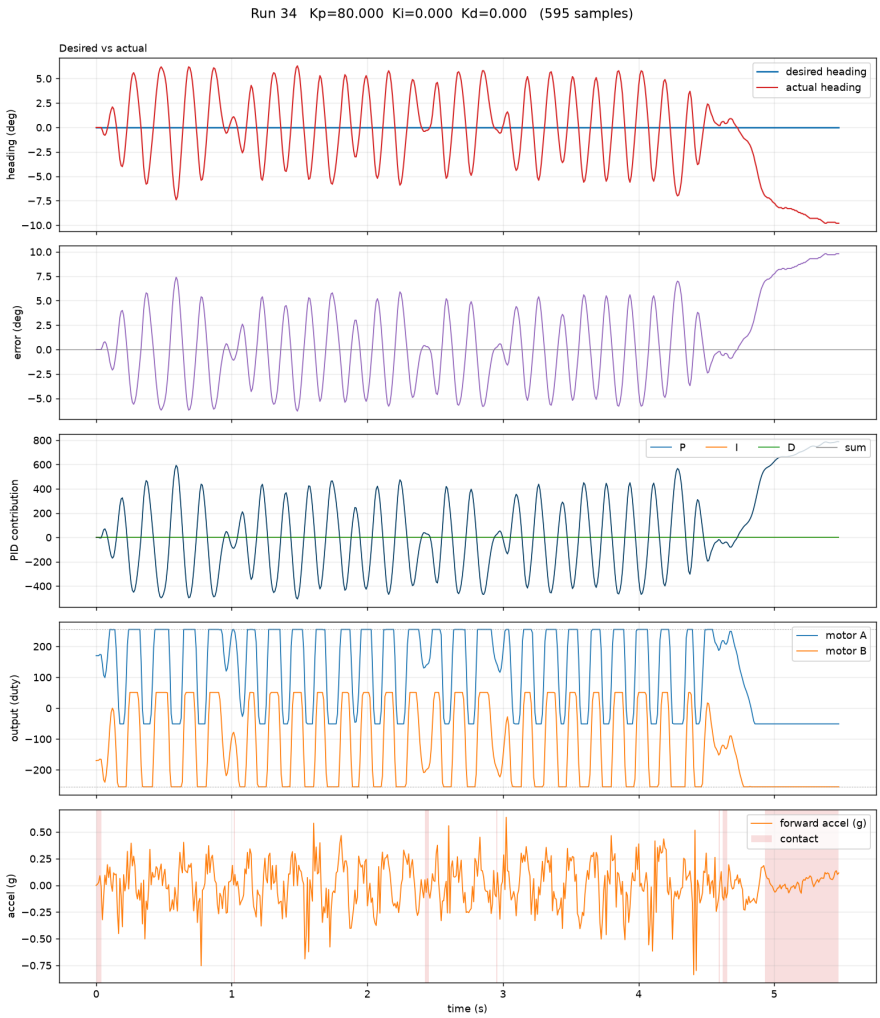


Figure 7.4.3: Testing if P implementation functions as it should. Higher P = more oscillations

The unstable readings at the end were due to the robot being picked up/ the button pressed to stop the reading.

7.5 Pushing objects

-> wheel slipping still the main issue, revisited wheel design. Can push up to 400g now, still not the desired 500g. -> could benefit from adding more weight near the front + a sharper wedge. Going to use sheet metal for this sharper wedge.

Chapter 8
Conclusions and Future Work

Appendix A

Planning

A.1 Gannt Chart: Progress to Date

Table A.1.1: Project schedule

Task	**Feb 2026**	**Mar 2026**	**Apr 2026**	**May 2026**	**Jun 2026**	**Jul 2026**	**Aug 2026**	**Sep 2026**	**Oct 2026**	**Nov 2026**
Review Literature										
Identify Design Approaches										
Compile Design Requirements										
Investigate Design Concepts										
Select Preferred Design										
Refine the Selected Design										
Simulate Components of Interest										
Finalize Design for Prototyping										
Manufacture Prototype										
Test Prototype										
Improve Prototype and Validate Performance										
Finalize Report										
Presentation										

Completed

Currently Busy

To do

*Note, the literature review will be updated and expanded in parallel with the design and testing phase

A.2 Budget and Procurement

Table A.2.1: Bill of Materials Procurement Status

Item	Qty	Cost	Total	Found	Email	Proc.	Rec.
Sensors							
ToF	5	R73.91	R369.55	✓	✓	✓	✓
IR	1	R36.00	R36.00	✓	✓	✓	✓
MPU-6050	1	R10.00	R10.00	✓	✓	✓	✓
Shipping + VAT		R208.32	R208.32				
		Sub-total	**R623.87**				
Microcontroller							
STM32F411 Black Pill	1	R158.00	R158.00	✓	✓	✓	✓
Shipping		–	–				
		Sub-total	**R158.00**				
Power System							
N20 DC Motor	2	R35.00	R70.00	✓	✓	✓	✓
Motor Driver	1	R206.00	R206.00	✓	✓	✓	✓
11.1–16 V LiPo Battery	1	R619.00	R619.00	✓	✓	✓	✓
Shipping		R0.00	R0.00				
		Sub-total	**R895.00**				
Miscellaneous							
Prototyping Board	1	R200	R200	✓	✓	✓	✓
DIP Switch	1	R10	R10	✓	✓	✓	✓
3D Printing Frame Parts	1	R1000	R1000	✓	✓	✓	✓
Power Switch	1	R10	R10	✓	✓	✓	✓
Wheel Material (Vytaflex)	1	R350	R350	✓	✓	✓	✓
Passive Components	1	R400	R400	✓	✓	✓	✓
		Sub-total	**R1970.00**				
		Grand Total	**R3646.87**				

*Man hours will be updated for a separate BOM. The current list shows the progress of procurement/ budget use.

Appendix B

Calculations

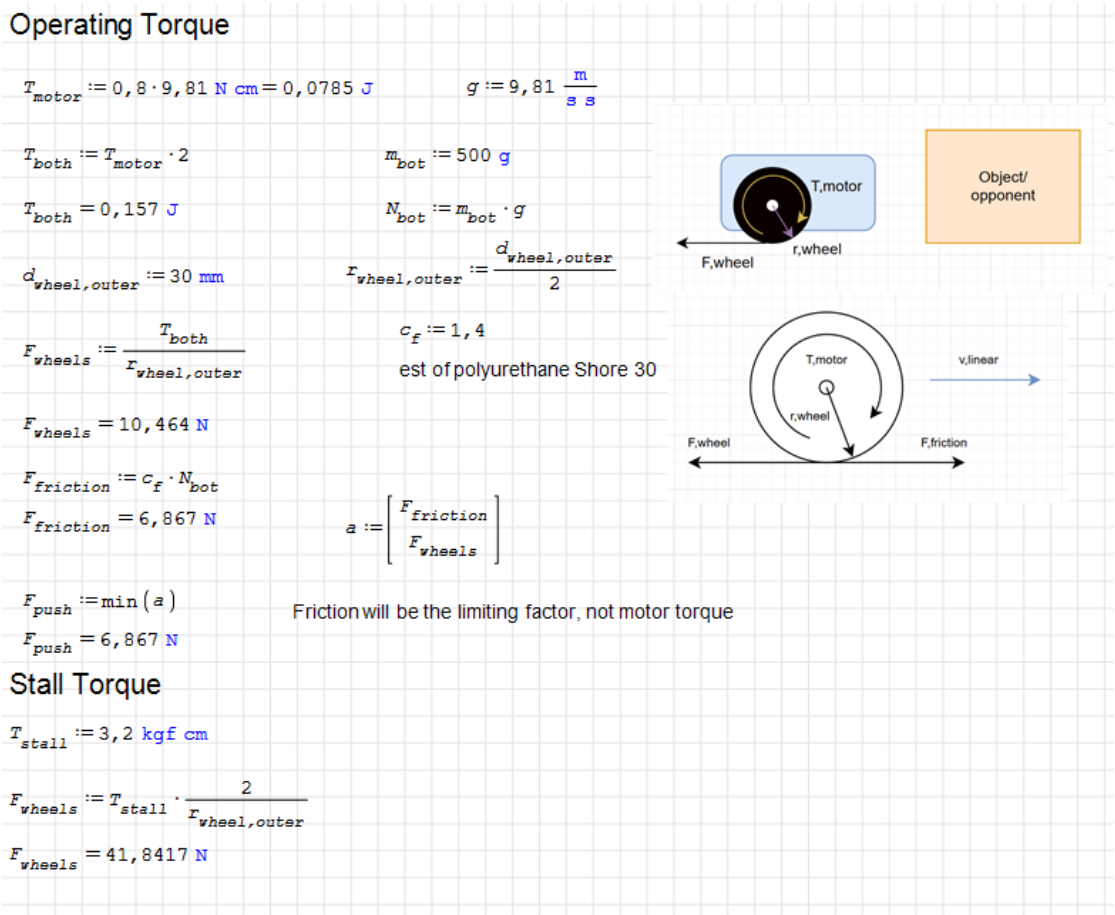


Figure B.0.1: Motor Torque and Wheel Material.

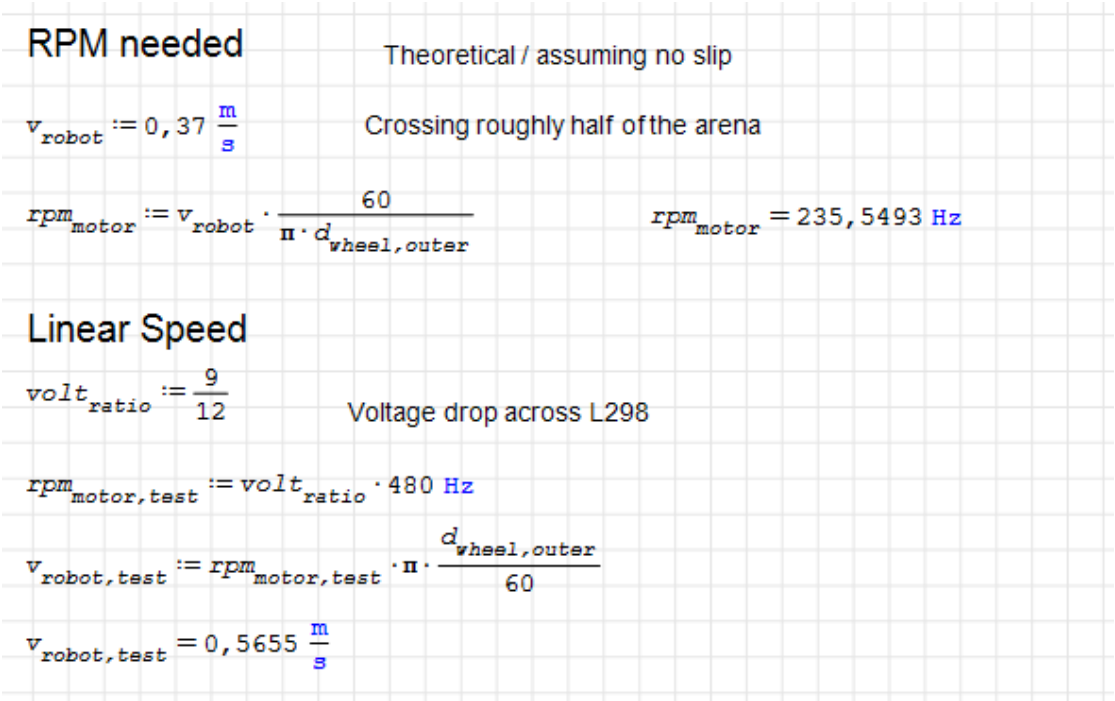


Figure B.0.2: Motor Selection Calculations.

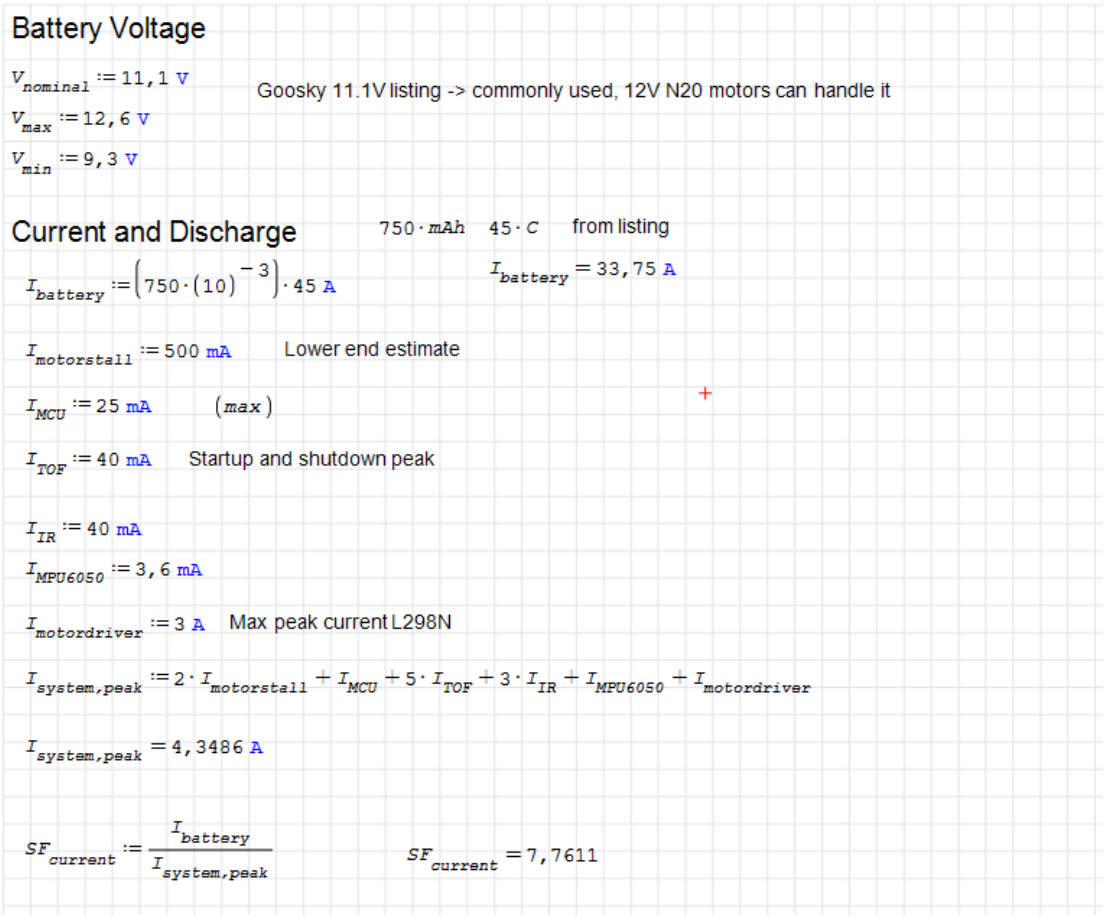


Figure B.0.3: Battery Calculations.

Appendix C

Schematic

C.1 KiCad Schematic

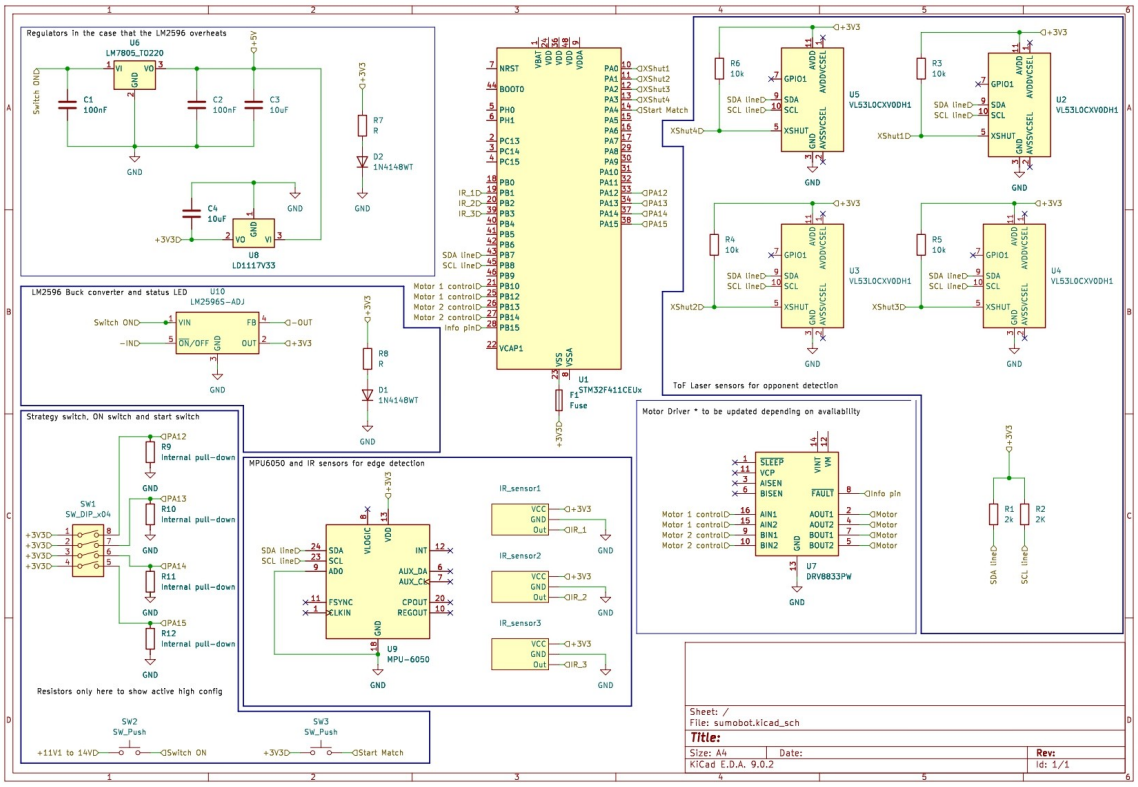


Figure C.1.1: KiCad schematic of the Mini Sumo robot.

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